

This project responds to an imagined request from the Montgomery County Department of Transportation and Ride On planning team. The agency has implemented Ride On Reimagined, a bus network redesign intended to modernize the county's transit network. Because bus network redesigns redistribute service across space, they raise an important equity question: did the redesign produce larger scheduled service gains in neighborhoods where residents are more likely to depend on transit?

The central question for this analysis is: Did Ride On Reimagined Phase 1 produce larger scheduled service gains in higher-transit-need Census tracts than in lower-transit-need tracts? The hypothesis is that higher-transit-need tracts received larger scheduled weekday service gains after Phase 1 implementation than lower-transit-need tracts. This is an empirical hypothesis rather than a normative claim. It does not ask what Ride On should have done; it tests what happened in the published schedule data.

The analysis uses three main data sources. First, it uses Ride On GTFS schedule feeds to measure planned transit service. The primary pre-implementation baseline is the January 2025 GTFS feed, active from January 12, 2025 to June 28, 2025. The primary post-implementation feed is the June 2025 GTFS feed, active from June 29, 2025 to September 6, 2025. This timing is important because June 29, 2025 is the official Phase 1 implementation date for Ride On Reimagined. Second, the project uses 2020-2024 ACS 5-year tract data to measure neighborhood transit need. Third, it uses 2024 TIGER/Line Census tract boundaries to connect schedule data and demographic indicators spatially.

The method has two linked parts. The first part measures transit need. I created a tract-level Transit Need Index using four ACS indicators: percent of households without vehicles, percent of people below poverty, percent age 65 and over, and percent disabled. These variables were standardized and averaged so tracts could be grouped into need quartiles from low to high. The second part measures scheduled service change. GTFS stop-level service metrics were calculated for each feed, including weekday scheduled trips, weekend scheduled trips, peak-period trips per hour, and span of service. Stops were then connected to Census tracts using 0.25-mile stop catchments. This creates a tract-level measure of nearby scheduled service for each feed.

After building the tract-level service panel, I compared scheduled weekday service change from January 2025 to June 2025 across transit need quartiles. This is the main pre/post Phase 1 test. I also used a May 2026 published GTFS feed as a sensitivity check, but not as the primary result, because that feed includes future scheduled dates as of May 2026. The 2026 comparison is therefore treated as planned schedule evidence rather than observed service performance.

The main result does not strongly support the hypothesis that higher-transit-need tracts received larger scheduled service gains. In the Phase 1 comparison, low-need tracts changed by about -3.0 accessible weekday trips on average, moderate-low need tracts gained about +66.0, moderate-high need tracts gained about +18.4, and high-need tracts changed by about -26.7. The statistical tests point in the same direction. The Welch t-test comparing high-need tracts to all other tracts is not statistically significant, and the Pearson and Spearman correlations between Transit Need Index and weekday service change are close to zero and not significant.

This conclusion does not mean that Ride On Reimagined failed as a transit redesign. The more careful interpretation is that, under this tract-level scheduled-service method, the largest average gains were not concentrated in the highest-transit-need quartile. The trend chart also shows that high-need tracts started with the highest average nearby scheduled weekday service before implementation. That makes the equity story more complicated. The issue is not simply whether high-need tracts had service; it is whether they received additional service gains after the redesign.

The maps help show why the result matters spatially. The Transit Need Index map identifies where higher-need tracts are located. The Phase 1 service-change map shows that scheduled service gains and losses were geographically uneven. Some tracts gained nearby scheduled trips, while others lost them or changed very little. The high-need watchlist map identifies tracts that combine high measured transit need with negative scheduled service change. These places should not be treated as automatic proof of harm, but they are important candidates for closer agency review. A planning agency could use these tracts to ask more detailed questions about route changes, stop access, transfers, destination access, and rider experience.

The broader implication is that smart-city and data science tools can make equity patterns visible, but they can also narrow the meaning of equity. GTFS and ACS data are useful because they are public, standardized, spatial, and reproducible. They make it possible to audit whether scheduled service changes align with demographic indicators of transit need. However, this approach also has limits. GTFS measures scheduled service, not actual service delivery. It does not show whether buses arrived on time, whether trips were missed, whether buses were crowded, whether riders felt safe, or whether routes connected residents to jobs, schools, health care, and groceries. ACS indicators also simplify need into measurable variables, even though community-defined need may include concerns that are not captured in the index.

Several methodological limitations follow from these choices. The 0.25-mile catchment is a simplified measure of stop proximity and does not account for sidewalks, street crossings, hills, barriers, or perceived safety. Census tracts are useful for analysis but do not perfectly represent neighborhoods or rider catchments. The Transit Need Index uses equal weighting across selected ACS variables, which is transparent but not the only possible weighting scheme. The project addresses these challenges by clearly limiting claims to scheduled service supply, using a valid pre/post implementation comparison, treating the 2026 feed only as a sensitivity check, and identifying specific tracts for follow-up rather than claiming that the index fully represents rider experience.

Overall, the analysis suggests that Ride On Reimagined Phase 1 did not clearly produce larger scheduled service gains in the county's highest-transit-need tracts. For a planning agency, the finding points to a need for deeper equity review rather than a simple judgment of success or failure. A next step would be to combine this scheduled-service audit with route-level analysis, destination accessibility measures, reliability data, and rider/community input. In that model, data science supports public accountability, but it does not replace the lived knowledge of transit users.